Hotel Booking Cancellation

Analysis and Risk Segmentation - Project Report

Scope	117,399 bookings (after cleaning) from a City Hotel and a Resort Hotel in Portugal, July 2015 - August 2017
Tools	Python (pandas, Google Colab), SQL (DuckDB), Tableau Public
Dashboard	Who Cancels More? Hotel Booking Risk Dashboard (Tableau Public, link in the README)

1. Executive summary

37.5% of bookings were canceled, which puts about **16.7M (39%) of potential revenue** at risk, or about 13.1M (31%) if non-refundable bookings are treated as retained revenue. The problem is concentrated: bookings made more than 90 days ahead, Online Travel Agents (Online TA) and Groups, and the City Hotel explain most of it. The data shows where the risk is, not what would change guest behavior, so the recommendations below are hypotheses to validate with an A/B test.

2. Business problem

A hospitality group operates two hotels in Portugal. Each cancellation leaves a room unsold, and a late one leaves little time to resell it. The team has no clear view of which bookings are truly at risk, so the same policy is applied to all of them.

Business question: Which booking profiles are most likely to cancel, and what policy (deposits, cancellation fees, controlled overbooking) can reduce or recover the revenue lost without discouraging guests from booking?

3. Data and preparation

Source: *Hotel Booking Demand* dataset (Antonio, Almeida and Nunes, 2019), 119,390 reservation-level records. Cleaning was applied on a working copy:

Rule	Rows removed	Rows remaining
Initial rows	-	119,390
Bookings with no guests	180	119,210
ADR <= 0	1,811	117,399
Zero nights	0	117,399

- The 715 zero-night bookings already had ADR <= 0, so the rules overlap.
- Bookings with no guests, zero nights or ADR <= 0 were treated as likely system errors (an assumption).
- Duplicates were kept: with no booking ID, identical rows may be different bookings.
- Revenue is defined as ADR x total nights (potential revenue). The cancellation rate stays at about 37.5% after cleaning.

4. Key findings

Lead time is the strongest signal. Cancellation rises from 9.7% for bookings made within a week to 57.2% for those made more than 180 days ahead. Bookings made more than 90 days ahead account for 58.7% of all cancellations.

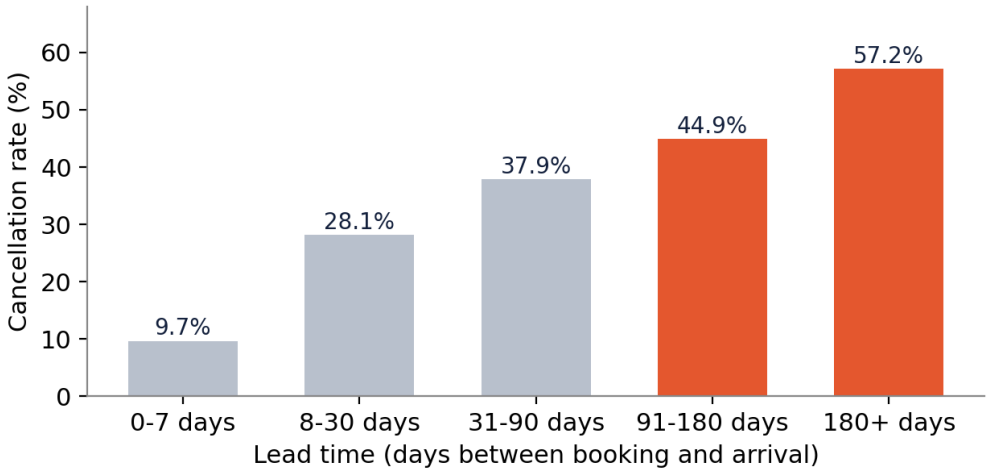


Figure 1. Cancellation rate by lead time group.

Online TA drives the revenue loss. Groups has the highest cancellation rate (61.7%), but Online TA accounts for 47.1% of cancellations and 61.1% of revenue lost, because of its volume and higher booking value. Together, Online TA and Groups account for 74.5% of cancellations.

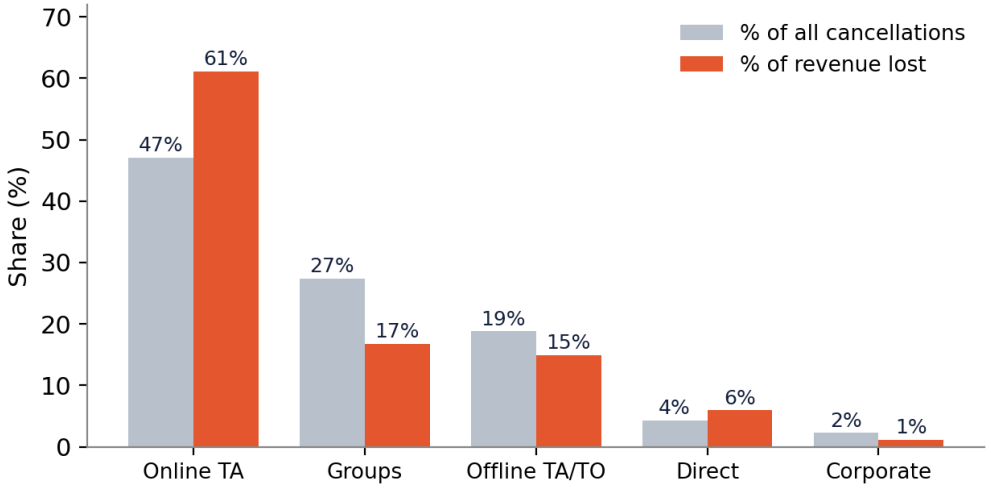


Figure 2. Share of cancellations and share of revenue lost, top five segments by cancellations.

The two hotels behave differently. The City Hotel cancels 42.2% of bookings versus 28.1% at the Resort Hotel. City peaks in April-June (about 45-47%); the Resort is lowest in winter (15-19%) and highest in summer (about 32-34%). July and August have three years of data and the other months two, so compare rates, not volumes.

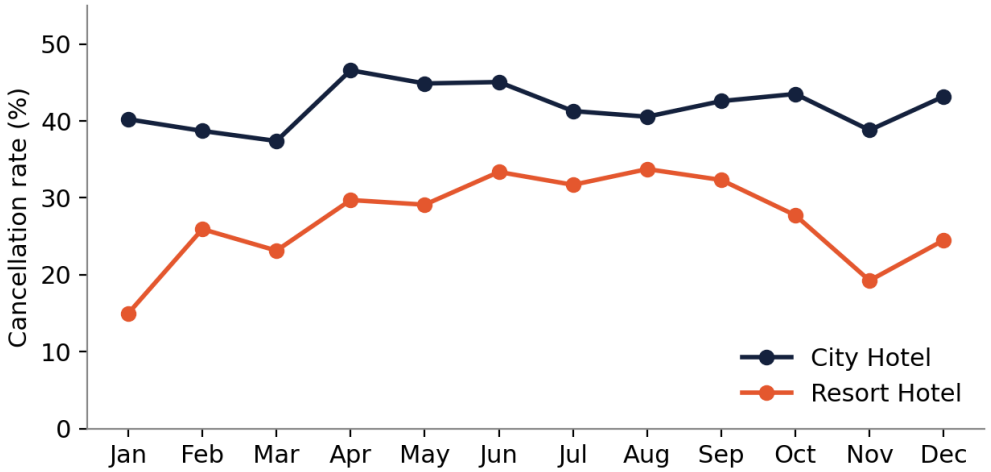


Figure 3. Cancellation rate by arrival month and hotel.

Other patterns

- **Customer type:** Transient customers account for 82.6% of cancellations. (customer_type "Group" is not the same as market_segment "Groups".)
- **Country:** Portugal is 40% of bookings and 62.1% of cancellations. Among No Deposit bookings it still cancels more (40.1% vs 23.4% elsewhere), so the gap is not explained by Non Refund alone.
- **Special requests:** 48.3% of bookings with no requests cancel versus 22.0% with one or more. This is an association: requests are usually made after booking.
- **Previous cancellations:** a real signal, with about 87% cancellation among No Deposit bookings with one previous cancellation (the raw 94.7% is inflated by Non Refund).

5. Revenue at stake

The dataset has no payment data, so two scenarios are shown. Scenario A assumes all canceled revenue is lost. Scenario B treats Non Refund cancellations as retained revenue. Even in the conservative scenario, close to a third of potential revenue is lost, almost all of it from No Deposit bookings.

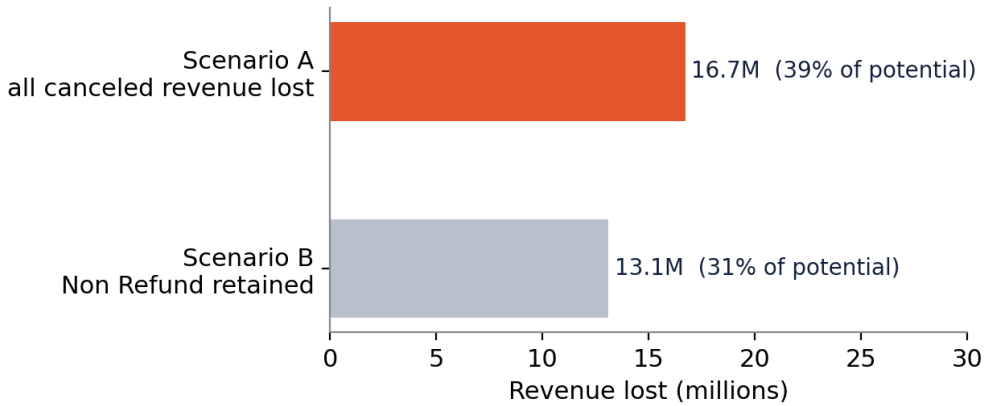


Figure 4. Revenue lost to cancellations under two assumptions.

6. Recommendations

These are hypotheses to validate before rolling out any policy change.

- **Target long lead times (90+ days):** test deposits, tiered cancellation fees or pre-arrival confirmations. These bookings concentrate 58.7% of cancellations.
- **Prioritize Online TA and Groups:** test stricter terms with Groups, and deposits or cancellation fees with Online TA, weighing the risk of losing bookings to competitors.
- **Use different policies per hotel and season:** City Hotel needs measures all year (strongest in spring); Resort Hotel mainly in summer.
- **Recover revenue from the highest-risk profiles:** a higher cancellation fee can recover part of the loss whether or not the room is resold.
- **Encourage personalization:** guests with special requests cancel far less, so inviting them to customize the stay (room, arrival time) may raise commitment.
- **Validate with an A/B test** before any wider rollout.

7. Limitations and next steps

- **Non Refund anomaly:** these bookings show about 99% cancellation, concentrated in Portugal, Groups and Offline TA/TO. With no payment data, it is unclear whether that revenue was retained or whether deposit type is recorded reliably. Next step: validate with the finance team how deposits are recorded and charged.
- **Pre-COVID data (2015-2017):** it may not reflect current booking behavior.
- **Associations, not causes:** findings describe patterns in historical data.
- **Next steps:** an explainable model (for example logistic regression) to score booking risk, and a risk-versus-impact prioritization matrix in the dashboard.